

SRS-Immune Fiber Attenuation Monitoring via Full-Band Power Aggregation in C+L Band Systems

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Abstract We propose a full-band power aggregation method for real-time fiber attenuation monitoring in C+L band systems to overcome stimulated Raman scattering (SRS) interference in single-channel monitoring. Validation on 237 production spans reduces the false alarm rate from 24.9% to 0% at a 3.5dB attenuation variation. ©2026 The Author(s)

Introduction

The growth of AI workloads is driving inter-datacenter traffic from the C-band into C+L band systems to double per-fiber capacity [1–3]. This upgrade raises the stakes for fiber fault management, as a single degradation now disrupts twice as many channels. Real-time attenuation monitoring is therefore critical for rapid fault detection and automatic amplifier optimization [4].

However, stimulated Raman scattering (SRS) transfers power from C-band to L-band channels [5], causing single-channel power readings to fluctuate with the total traffic load rather than reflecting the true fiber condition. When a fiber cut or amplifier shutdown on one span eliminates C+L signals, the SRS-induced offset vanishes on all other spans, triggering widespread false alarms on intact fibers.

We propose a full-band power aggregation method that sums C-band, L-band, and OSC powers to exploit the energy-conserving nature of SRS, inherently cancelling inter-band redistribution. The method requires no additional hardware beyond existing telemetry infrastructure. Validation on 237 non-fault spans from Tencent's production network demonstrates that the method reduces the false alarm rate from 35.9% (single-channel) to 0.4% at 3 dB deviation, and achieves 0% at 3.5 dB.

Online Fiber Monitoring

Conventional fiber characterization techniques—OTDR, OFDR, BOTDR—require either service interruption or a dedicated measurement channel, making them unsuitable for live traffic monitoring. Among in-service approaches, OTDR—despite advances in coded, coherent, and AI-assisted variants [6–9]—still requires injecting test pulses that risk co-channel interference, and its minute-scale measurement cycle precludes real-time fault detection. In practice, OTDR serves primarily for *post-event* fault localization rather than continuous monitoring.

To overcome these limitations, we deploy a hybrid

monitoring framework in Tencent open optical platform (TOOP) that couples distributed power monitoring with anomaly-triggered OTDR: second-scale power analysis detects fiber degradation in real time, while OTDR scans are invoked on-demand to pinpoint fault locations. In C-band-only deployments, the telemetry platform [10] collects OSC power at both ends of each span; deviations from the commissioning baseline trigger the automatic amplifier configuration system [4] for proactive compensation or operator alerts. However, extending this single-channel approach to C+L band systems is compromised by SRS, as discussed in Section 1. To address this, we propose a full-band fiber attenuation measurement method described in the following section.

Real-Time Fiber Attenuation Measurement

Figure 1 illustrates the architecture of OpTel, our vendor-agnostic optical telemetry platform deployed across Tencent's C+L band transport network [10]. In a typical multi-span link, the optical signal traverses N amplified spans, each consisting of a booster amplifier (BA), a fiber segment, and a preamplifier (PA), with dedicated amplification chains for the C-band and L-band. An optical supervisory channel (OSC) at 1625 nm co-propagates with the payload signal for out-of-band management.

OpTel collects optical performance metrics at one-second granularity [10], including booster/preamplifier port powers ($P_{\text{BA},\text{out}}$, $P_{\text{BA},\text{in}}$, $P_{\text{PA},\text{in}}$, $P_{\text{PA},\text{out}}$), VOA settings for each amplifier stage, and bidirectional OSC power levels. These observables serve as inputs to the fiber attenuation estimation described below.

The fundamental principle is the span-level power budget: the difference between total launched and received power reflects the aggregate propagation loss. The fiber attenuation is:

$$\mathcal{L}_{\text{fiber}} = P_{\text{tx}} - P_{\text{rx}} \quad (1)$$

where P_{tx} and P_{rx} denote the total optical power launched into and received from the fiber (dBm).

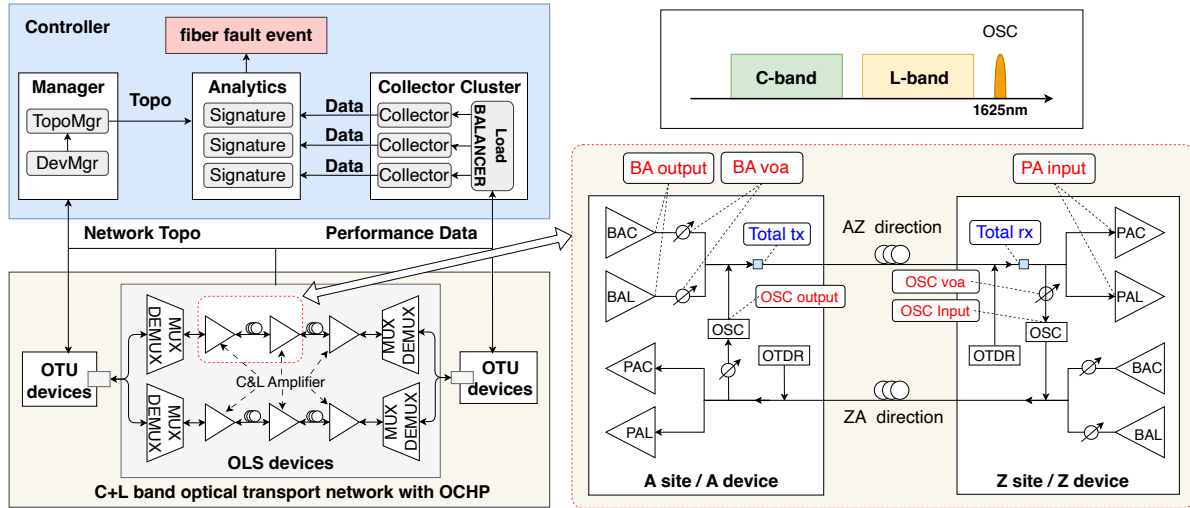


Fig. 1: System architecture of the proposed full-band fiber fault monitoring.

The key challenge lies in accurately estimating these quantities from telemetry observables while accounting for VOA placement asymmetries.

Single-Channel (SC) Method. In C-band-only systems, the wavelength-dependent attenuation variation across the C-band is negligible (<0.1 dB for G.652 fiber), so the more stable OSC channel suffices for attenuation estimation:

$$\mathcal{L}_{SC}^{az} = P_{OSC,out}^a - P_{OSC,in}^z - A_{OSC,VOA}^z \quad (2)$$

where $P_{OSC,out}^a$ is the OSC launch power at site A, $P_{OSC,in}^z$ is the received OSC power at site Z, and $A_{OSC,VOA}^z$ is the ingress VOA attenuation at site Z.

Full-Band (FB) Method. In C+L band systems, SRS induces inter-band power transfer from C-band to L-band [5], rendering single-channel measurements unreliable. Since SRS is an energy-conserving process, we propose aggregating the optical power across all co-propagating channels—C-band, L-band, and OSC—to cancel SRS-induced redistribution. The aggregate transmit power is:

$$P_{tx}^{az} = 10 \log_{10}(p_{C,out}^{az} + p_{L,out}^{az} + p_{OSC,out}^{az}) \quad (3)$$

where the linear-domain component powers are:

$$p_{C,out}^{az} = 10^{(P_{BAC,out} - A_{BAC,VOA})/10} \quad (4)$$

$$p_{L,out}^{az} = 10^{(P_{BAL,out} - A_{BAL,VOA})/10} \quad (5)$$

$$p_{OSC,out}^{az} = 10^{P_{OSC,out}/10} \quad (6)$$

Here $A_{BAC,VOA}$ and $A_{BAL,VOA}$ are the egress VOA attenuations at the C-band and L-band booster outputs, which must be subtracted to recover the actual fiber launch power.

Analogously, the aggregate receive power P_{rx}^{az} accounts for the ingress VOA at the OSC receiver:

$$P_{rx}^{az} = 10 \log_{10}(p_{C,in}^{az} + p_{L,in}^{az} + p_{OSC,in}^{az}) \quad (7)$$

where:

$$p_{C,in}^{az} = 10^{P_{PAC,in}/10} \quad (8)$$

$$p_{L,in}^{az} = 10^{P_{PAL,in}/10} \quad (9)$$

$$p_{OSC,in}^{az} = 10^{(P_{OSC,in} + A_{OSC,VOA})/10} \quad (10)$$

The $A_{OSC,VOA}$ term compensates for the ingress VOA to recover the fiber-end OSC power. The full-band fiber attenuation is then:

$$\mathcal{L}_{FB}^{az} = P_{tx}^{az} - P_{rx}^{az} \quad (11)$$

Since SRS conserves total optical power, the full-band summation inherently cancels SRS-induced redistribution, yielding a robust estimate of the true fiber attenuation regardless of the traffic load or SRS strength.

Experimental Results

We validate the proposed method using data collected from Tencent's production C+L band optical transport network at one-second granularity. The dataset comprises two categories of service interruption events across multi-span links (4-span, 7-span, and 9-span): (i) 7 fiber cut events involving 99 non-fault spans, and (ii) 16 upstream amplifier shutdown events involving 138 spans, yielding 237 non-fault span measurements in total. In both categories, the loss of C+L signal power eliminates the SRS-induced Raman gain, triggering power redistribution on all remaining spans.

We define the *attenuation deviation* Δ as the change in measured fiber attenuation between the normal and interrupted states:

$$\Delta_{SC} = \mathcal{L}_{SC}^{int} - \mathcal{L}_{SC}^{norm} \quad (12)$$

$$\Delta_{FB} = \mathcal{L}_{FB}^{int} - \mathcal{L}_{FB}^{norm} \quad (13)$$

For a non-fault span whose physical fiber is intact, the ideal Δ is zero. A false alarm occurs when $|\Delta|$ exceeds the detection threshold Δ_{th} .

Representative Case Analysis. Table 1 presents a representative 9-span system where Span 9 experiences a fiber cut. The FB–SC column quantifies the SRS-induced power transfer: under normal conditions, SRS makes the SC reading 0.51–3.54 dB *lower* than the FB reading. After the cut eliminates all downstream C+L power, this SRS offset vanishes and the SC attenuation abruptly increases. The decomposition identity $\Delta_{SC} = (FB - SC) + \Delta_{FB}$ holds across all spans, confirming that the SC anomaly comprises the vanished SRS offset plus a small residual component. At a 3.5 dB threshold, 3 of 8 normal spans trigger SC false alarms (Spans 1, 2, 5), while the Δ_{FB} of these spans still within 0.56–1.69 dB.

Tab. 1: 9-Span system with a fiber cut event on span 9

Span	Normal (dB)			Interrupted (dB)		
	SC	FB	FB–SC	SC(=FB)	Δ_{SC}	Δ_{FB}
1	15.19	18.73	3.54	19.65	+4.46	+0.92
2	17.15	19.48	2.33	21.17	+4.02	+1.69
3	23.27	24.73	1.46	26.33	+3.06	+1.60
4	16.21	18.47	2.26	19.03	+2.82	+0.56
5	20.76	22.82	2.06	24.27	+3.51	+1.45
6	24.60	25.20	0.60	26.45	+1.85	+1.25
7	23.63	25.61	1.98	27.05	+3.43	+1.44
8	19.67	20.18	0.51	21.11	+1.44	+0.93
9	20.95	21.69	0.74	44.65	+23.70	+22.96

After the cut, only OSC remains; SC and FB readings converge.

Aggregate False Alarm Statistics. Table 2 summarizes the false alarm rates across all 237 non-fault spans at various thresholds. The SC method exhibits a 35.9% false alarm rate at 3 dB, whereas the FB method reduces this to 0.4% (a single borderline case). At 3.5 dB, the FB method achieves zero false alarms while the SC method still triggers on 24.9% of spans. Notably, amplifier shutdown events produce higher SC false alarm rates (41.3%) than fiber cuts (28.3%), because the complete loss of C+L signals affects *all* spans simultaneously rather than only those downstream of a fault.

Tab. 2: False alarm rates across 237 non-fault spans

Δ_{th} (dB)	SC Method		FB Method	
	Count	Rate	Count	Rate
2.0	125/237	52.7%	4/237	1.7%
2.5	98/237	41.4%	1/237	0.4%
3.0	85/237	35.9%	1/237	0.4%
3.5	59/237	24.9%	0/237	0.0%

Attenuation-Dependent SC Bias. Figure 2 plots Δ_{SC} and Δ_{FB} against the normal-state fiber attenuation for all non-fault spans. Linear regression yields $\Delta_{SC} = -0.119\mathcal{L}_{fiber} + 4.74$ and $\Delta_{FB} = 0.030\mathcal{L}_{fiber} - 0.26$. The negative slope of Δ_{SC} indicates that lower-attenuation spans suffer larger SRS-induced bias, because they carry higher residual C+L power at the fiber output, resulting in stronger SRS interaction. In the 19–20 dB bin, the average $|\Delta_{SC}|$ reaches 3.09 dB (72.4% false alarm rate), whereas in the 25–27 dB bin it drops to 1.43 dB (7.9%). In contrast, Δ_{FB}

exhibits a near-zero slope, with $|\Delta_{FB}|$ remaining in 0.25–0.63 dB across all attenuation bins, confirming the immunity of the full-band method to attenuation-dependent SRS bias.

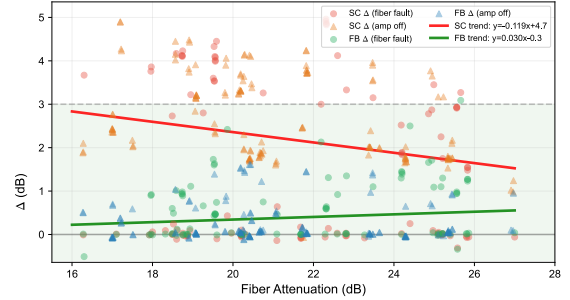


Fig. 2: Δ_{SC} and Δ_{FB} versus fiber attenuation for all 237 non-fault spans. The shaded band denotes the ± 3 dB alarm zone. Trend lines are obtained by linear regression.

Physical Interpretation. The small residual Δ_{FB} (mean 0.55 dB, std 0.69 dB) arises from wavelength-dependent fiber loss: after an interruption eliminates C+L channels, only the OSC signal at 1625 nm remains, whose attenuation coefficient (0.23 dB/km) exceeds the power-weighted average of the original C+L ensemble (0.20 dB/km), producing a modest upward offset [11]. The sign of Δ_{SC} is also consistent with SRS theory: 86.5% of values are positive, consistent with the fact that SRS normally transfers power from shorter (C-band) to longer (L-band and OSC) wavelengths, reducing the apparent OSC-measured span loss. Once the C+L signals vanish, this SRS “subsidy” disappears and the SC reading reverts to its true (higher) value.

Conclusions

We proposed and validated a full-band power aggregation method for real-time fiber attenuation measurement in C+L band systems. By summing all co-propagating channel powers, the method inherently cancels SRS-induced redistribution that corrupts single-channel measurements. Validation on 237 non-fault spans shows that the FB method reduces the false alarm rate from 35.9% to 0.4% at 3 dB attenuation deviation and achieves 0% at 3.5 dB. The attenuation-dependent analysis further confirms that SC bias worsens on lower-loss spans, while FB measurements remain stable across all ranges. The method requires no additional hardware and has been deployed in Tencent open optical platform for continuous fiber health monitoring.

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